

Gradient Descent

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Gradient descent

Consider unconstrained, smooth convex optimization

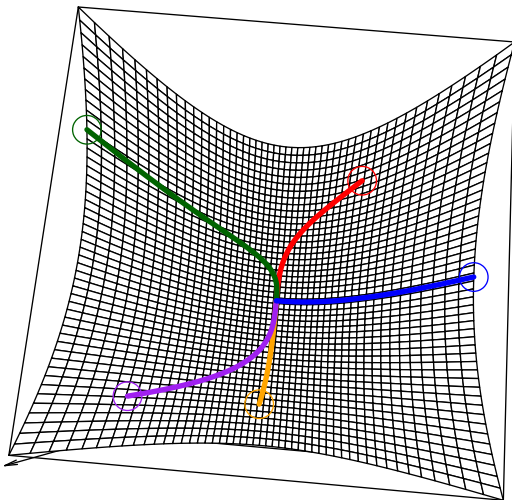
$$\min_x f(x)$$

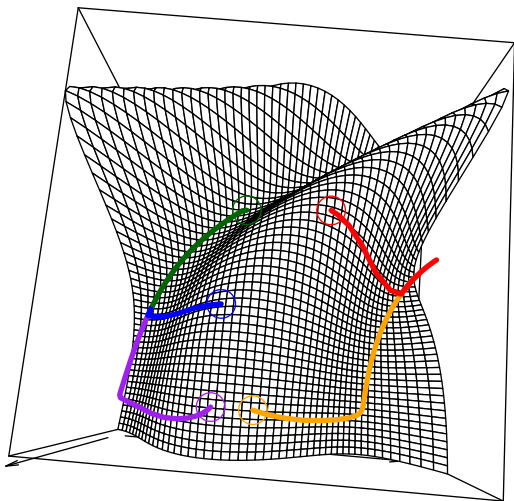
with convex and differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Denote the optimal value by $f^* = \min_x f(x)$ and a solution by x^* .

Gradient descent: choose initial point $x^{(0)} \in \mathbb{R}^n$, repeat:

$$x^{(k)} = x^{(k-1)} - t_k \cdot \nabla f(x^{(k-1)}), \quad k = 1, 2, 3, \dots$$

Stop at some point.





Gradient descent interpretation

At each iteration, consider the expansion

$$f(y) \approx f(x) + \nabla f(x)^T (y - x) + \frac{1}{2t} \|y - x\|_2^2.$$

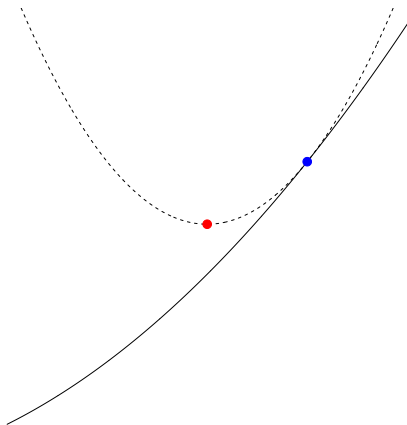
Quadratic approximation, replacing usual Hessian $\nabla^2 f(x)$ by $\frac{1}{t}I$.

$$\begin{array}{ll} f(x) + \nabla f(x)^T (y - x) & \text{linear approximation to } f \\ \frac{1}{2t} \|y - x\|_2^2 & \text{proximity term to } x, \text{ with weight } 1/2t \end{array}$$

Choose next point $y = x^+$ to minimize quadratic approximation

$$x^+ = x - t \nabla f(x).$$

Gradient descent interpretation



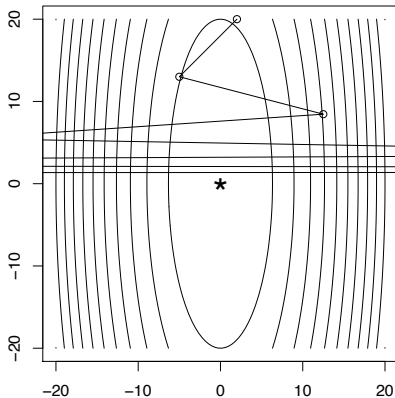
Blue point is x , red point is

$$x^* = \operatorname{argmin}_y f(x) + \nabla f(x)^T (y - x) + \frac{1}{2t} \|y - x\|_2^2$$

- ▶ How to choose step sizes
- ▶ Convergence analysis
- ▶ Nonconvex functions
- ▶ Gradient boosting

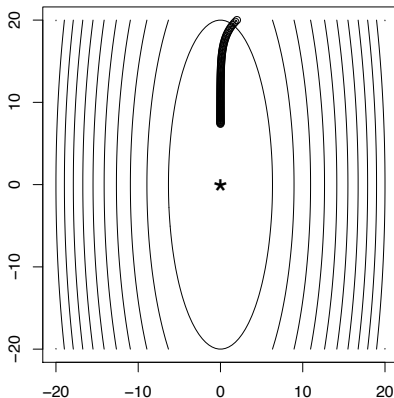
Fixed step size

Simply take $t_k = t$ for all $k = 1, 2, 3, \dots$, can **diverge** if t is too big.
Consider $f(x) = (10x_1^2 + x_2^2)/2$, gradient descent after 8 steps:



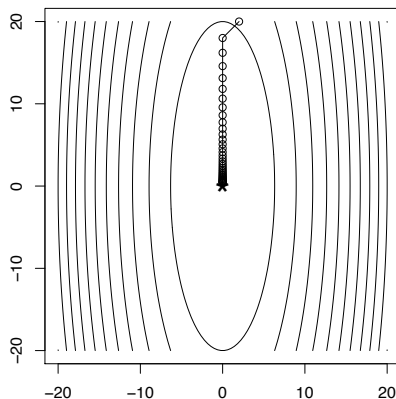
Fixed step size

Can be **slow** if t is too small. Same example, gradient descent after 100 steps:



Fixed step size

Converges nicely when t is “just right”. Same example, 40 steps:



Convergence analysis later will give us a precise idea of “just right”.

Backtracking line search

One way to adaptively choose the step size is to use **backtracking line search**:

- ▶ First fix parameters $0 < \beta < 1$ and $0 < \alpha \leq 1/2$.
- ▶ At each iteration, start with $t = t_{\text{init}}$, and while

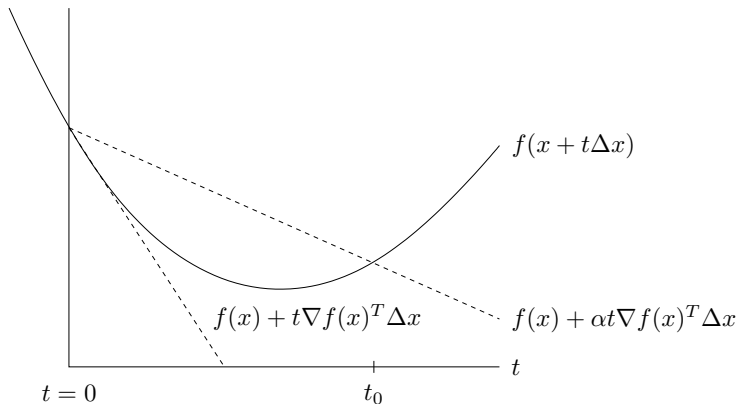
$$f(x - t\nabla f(x)) > f(x) - \alpha t \|\nabla f(x)\|_2^2$$

shrink $t = \beta t$. Else perform gradient descent update

$$x^+ = x - t\nabla f(x).$$

Simple and tends to work well in practice (further simplification: just take $\alpha = 1/2$).

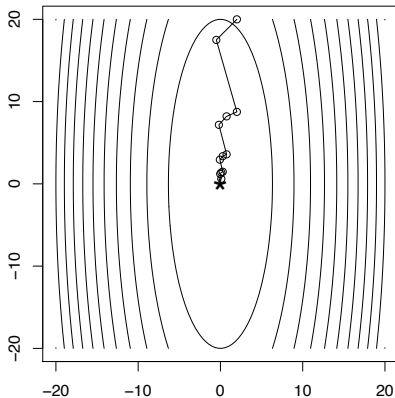
Backtracking interpretation



For us $\Delta x = -\nabla f(x)$

Backtracking line search

Setting $\alpha = \beta = 0.5$, backtracking picks up roughly the **right step size** (12 outer steps, 40 steps total).



Exact line search

We could also choose step to do the best we can along direction of negative gradient, called **exact line search**:

$$t = \operatorname{argmin}_{s \geq 0} f(x - s \nabla f(x)).$$

Usually not possible to do this minimization exactly.

Approximations to exact line search are typically not as efficient as backtracking and it's typically not worth it.

Convergence analysis

Assume that $f : \mathbb{R}^n \rightarrow \mathbb{R}$ convex and differentiable and additionally

$$\|\nabla f(x) - \nabla f(y)\|_2 \leq L \|x - y\|_2 \quad \text{for any } x, y$$

i.e., ∇f is Lipschitz continuous with constant $L > 0$.

Theorem

Gradient descent with fixed step size $t \leq 1/L$ satisfies

$$f(x^{(k)}) - f^* \leq \frac{1}{2tk} \|x^{(0)} - x^*\|_2^2$$

and same result holds for backtracking with t replaced by β/L .

We say gradient descent has convergence rate $O(1/k)$, i.e., it finds ε -suboptimal point in $O(1/\varepsilon)$ iterations.

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Slide 20-25 in <http://www.seas.ucla.edu/~vandenbe/236C/lectures/gradient.pdf>



Convergence under strong convexity

Reminder: **strong convexity** of f means $f(x) - \frac{m}{2} \|x\|_2^2$ is convex for some $m > 0$.

Assuming Lipschitz gradient as before and also strong convexity:

Theorem

Gradient descent with fixed step size $t \leq 2/(m + L)$ or with backtracking line search satisfies

$$f(x^{(k)}) - f^* \leq c^k \frac{L}{2} \|x^{(0)} - x^*\|_2^2$$

where $0 < c < 1$.

Rate under strong convexity is $O(c^k)$, exponentially fast, i.e., we find ϵ -suboptimal point in $O(\log(1/\epsilon))$ iterations.

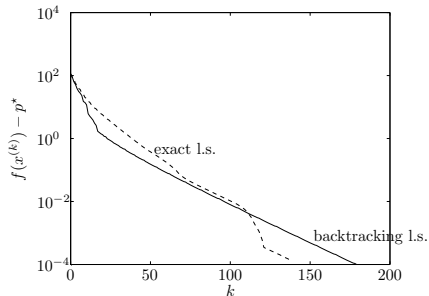
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Convergence rate

Called **linear convergence**, because looks linear on a semi-log plot.



(From B & V page 487)

Important note: contraction factor c in rate depends adversely on condition number L/m : higher condition number $\Rightarrow$ slower rate.

Affects not only our upper bound... very apparent in practice too.

A look at the conditions

A look at the conditions for a simple problem, $f(\beta) = \frac{1}{2} \|y - X\beta\|_2^2$.

Lipschitz continuity of ∇f :

- ▶ This mean $\nabla^2 f(\beta) \preceq LI$.
- ▶ As $\nabla^2 f(\beta) = X^T X$, we have $L = \sigma_{\max}(X^T X)$.

Strong convexity of f :

- ▶ This mean $\nabla^2 f(\beta) \succeq ml$.
- ▶ As $\nabla^2 f(\beta) = X^T X$, we have $m = \sigma_{\min}(X^T X)$.
- ▶ If X is wide (i.e., X is $n \times p$ with $p > n$), then $\sigma_{\min}(X^T X) = 0$, and f can't be strongly convex.
- ▶ Even if $\sigma_{\min}(X^T X) > 0$, can have a very large condition number $L/m = \sigma_{\max}(X^T X)/\sigma_{\min}(X^T X)$.

Stopping rule: stop when $\|\nabla f(x)\|_2$ is small

- ▶ Recall $\nabla f(x^*) = 0$ at solution x^*
- ▶ If f is strongly convex with parameter m , then

$$\|\nabla f(x)\|_2 \leq \sqrt{2m\varepsilon} \implies f(x) - f^* \leq \varepsilon.$$

Pros and cons of gradient descent:

- ▶ Pro: simple idea, and each iteration is cheap (usually)
- ▶ Pro: fast for well-conditioned, strongly convex problems
- ▶ Con: can often be slow, because many interesting problems aren't strongly convex or well-conditioned
- ▶ Con: can't handle nondifferentiable functions.

Can we do better?

Gradient descent has $O(1/\varepsilon)$ convergence rate over problem class of convex, differentiable functions with Lipschitz gradients.

First-order method: iterative method, which updates $x^{(k)}$ in $x^{(0)} + \text{span}\{\nabla f(x^{(0)}), \nabla f(x^{(1)}), \dots, \nabla f(x^{(k-1)})\}$.

Theorem (Nesterov)

For any $k \leq (n-1)/2$ and any starting point $x^{(0)}$, there is a function f in the problem class such that any first-order method satisfies

$$f(x^{(k)}) - f^* \geq \frac{3L \|x^{(0)} - x^*\|_2^2}{32(k+1)^2}.$$

Can attain rate $O(1/k^2)$, or $O(1/\sqrt{\varepsilon})$? Answer: **yes** (we'll see)!

What about nonconvex functions?

Assume f is differentiable with Lipschitz gradient as before, but now **nonconvex**. Asking for optimality is too much. So we'll settle for x such that $\|\nabla f(x)\|_2 \leq \varepsilon$, called **ε -stationarity**.

Theorem

Gradient descent with fixed step size $t \leq 1/L$ satisfies

$$\min_{i=0,\dots,k} \|\nabla f(x^{(i)})\|_2 \leq \sqrt{\frac{2(f(x^0) - f^*)}{t(k+1)}}.$$

Thus gradient descent has rate $O(1/\sqrt{k})$, or $O(1/\varepsilon^2)$, even in the nonconvex case for finding stationary points.

This rate **cannot be improved** (over class of differentiable functions with Lipschitz gradients) by any deterministic algorithm¹.

¹Carmon et al. (2017), "Lower bounds for finding stationary points I"

Key steps:

- ▶ ∇f Lipschitz with constant L means

$$f(y) \leq f(x) + \nabla f(x)^T(y - x) + \frac{L}{2} \|y - x\|_2^2, \quad \forall x, y.$$

- ▶ Plugging in $y = x^+ = x - t\nabla f(x)$,

$$f(x^+) \leq f(x) - \left(1 - \frac{Lt}{2}\right) t \|\nabla f(x)\|_2^2.$$

- ▶ Taking $0 < t \leq 1/L$, and rearranging,

$$\|\nabla f(x)\|_2^2 \leq \frac{2}{t}(f(x) - f(x^+)).$$

- ▶ Summing over iterations

$$\sum_{i=0}^k \left\| \nabla f(x^{(i)}) \right\|_2^2 \leq \frac{2}{t}(f(x^{(0)}) - f(x^{(k+1)})) \leq \frac{2}{t}(f(x^{(0)}) - f^*).$$

- ▶ Lower bound sum by $(k+1) \min_{i=0,1,\dots} \|\nabla f(x^{(i)})\|_2^2$, conclude.

References and further reading

-  S. Boyd and L. Vandenberghe (2004), *Convex optimization*, Chapter 9
-  T. Hastie, R. Tibshirani and J. Friedman (2009), *The elements of statistical learning*, Chapters 10 and 16
-  Y. Nesterov (1998), *Introductory lectures on convex optimization: a basic course*, Chapter 1
-  L. Vandenberghe, *Lecture notes for EE 236C*, UCLA, Spring 2011-2012